ref input, SimPoint estimates — small-fa

L1D 1 sets x 64, L2 1 sets x 512 · metric schema 9 · gem5 TimingSimpleCPU · 4 November 2026

36 of 36 runs completed, 0 running, 0 pending; **0 remaining**.

Program (does not depend on the cache)

run	configs done	spill/ROI	spill ld/load	spill st/store	(spill+csave)/ROI	(spill+csave) ld/load	(spill+csave) st/store
cactuBSSN	1/1	20.65%	47.95%	56.30%	20.67%	48.00%	56.39%
parest	1/1	0.16%	0.43%	0.62%	0.64%	1.19%	7.86%
deepsjeng	1/1	2.92%	9.17%	16.25%	8.11%	23.58%	49.72%
leela	1/1	0.38%	1.36%	1.80%	6.07%	15.90%	45.53%
namd	1/1	3.27%	8.21%	17.87%	3.27%	8.22%	17.90%
lbm	1/1	10.48%	32.28%	44.97%	10.48%	32.28%	44.97%
nab	1/1	5.20%	15.37%	34.02%	6.75%	18.74%	49.89%
xz_cld	1/1	3.38%	12.11%	22.88%	5.08%	17.35%	36.71%
xz_cpu2006docs	1/1	1.47%	5.78%	23.30%	2.03%	7.38%	37.90%
xz_combined	1/1	2.85%	10.66%	26.32%	4.17%	14.65%	42.32%
imagick	1/1	0.23%	0.12%	3.44%	5.33%	12.37%	46.11%
blender	1/1	0.38%	0.88%	3.23%	3.72%	6.96%	42.51%
povray	1/1	0.27%	0.46%	0.95%	14.26%	22.18%	55.24%
omnetpp	1/1	0.00%	0.00%	0.00%	12.75%	24.67%	48.13%
xalancbmk	1/1	0.13%	0.26%	0.80%	4.92%	8.32%	39.84%
x264_pass1	1/1	7.07%	15.02%	38.53%	10.18%	21.51%	55.90%
x264_pass2	1/1	6.25%	13.38%	40.16%	8.63%	18.23%	56.21%
x264_seek	1/1	6.41%	13.66%	41.07%	8.82%	18.56%	57.26%
gcc_pp_O3	1/1	1.23%	2.73%	4.48%	16.24%	33.66%	64.23%
gcc_pp_O2	1/1	1.25%	2.75%	4.47%	16.60%	34.40%	63.73%
gcc_smaller	1/1	1.09%	2.19%	4.61%	13.76%	25.29%	64.09%
gcc_ref32_O5	1/1	0.91%	1.84%	3.40%	15.53%	30.63%	59.73%
gcc_ref32_O3	1/1	0.68%	1.85%	1.06%	11.25%	27.66%	19.94%
mcf	1/1	1.40%	2.06%	7.74%	2.33%	3.60%	12.43%
perlbench_checkspam	1/1	7.98%	9.83%	29.42%	22.33%	36.24%	69.19%
perlbench_diffmail	1/1	1.96%	4.19%	4.64%	17.81%	31.25%	54.38%
perlbench_splitmail	1/1	4.16%	4.19%	15.82%	23.27%	35.82%	68.04%
bwaves_1	1/1	2.51%	6.24%	18.43%	2.51%	6.24%	18.43%
bwaves_2	1/1	2.59%	6.43%	18.99%	2.59%	6.43%	18.99%
bwaves_3	1/1	2.51%	6.23%	18.45%	2.51%	6.23%	18.45%
bwaves_4	1/1	2.56%	6.32%	19.44%	2.56%	6.32%	19.44%
exchange2	1/1	5.23%	10.55%	12.68%	6.08%	12.17%	14.84%
fotonik3d	1/1	3.36%	9.32%	9.25%	3.38%	9.37%	9.40%
roms	1/1	4.37%	13.83%	8.97%	4.44%	13.96%	9.48%
wrf	1/1	0.91%	3.38%	2.59%	1.52%	4.76%	7.57%
cam4	1/1	3.14%	11.02%	11.03%	4.99%	15.77%	21.23%
pooled		3.78%	9.06%	17.53%	7.49%	16.44%	39.26%
mean		3.31%	8.11%	15.78%	8.49%	18.33%	39.00%

Miss rates

run	state	L1D	spill ld L1	other ld L1	spill st L1	other st L1	L2	spill ld L2	spill st L2
cactuBSSN	done	15.37%	4.99%	32.48%	4.97%	12.55%	61.01%	12.41%	4.79%
parest	done	9.78%	1.18%	10.35%	1.28%	4.82%	75.05%	4.89%	23.16%
deepsjeng	done	7.89%	1.59%	10.13%	7.48%	3.94%	5.04%	2.01%	11.57%
leela	done	4.02%	0.06%	4.59%	0.00%	2.53%	28.13%	0.40%	20.61%
namd	done	11.54%	2.41%	14.70%	1.29%	2.41%	27.21%	7.40%	15.85%
lbm	done	17.10%	0.00%	6.08%	0.02%	73.68%	99.98%	100.00%	54.85%
nab	done	3.91%	0.08%	5.49%	0.25%	0.42%	82.17%	14.86%	4.12%
xz_cld	done	10.09%	0.50%	15.00%	0.27%	2.20%	36.73%	0.10%	1.48%
xz_cpu2006docs	done	9.30%	1.37%	10.33%	0.97%	5.74%	39.57%	7.56%	12.89%
xz_combined	done	9.69%	0.76%	12.63%	0.47%	3.86%	24.09%	0.00%	3.30%
imagick	done	2.22%	3.94%	2.83%	0.01%	0.12%	100.00%	57.17%	54.03%
blender	done	8.01%	2.59%	8.78%	1.35%	3.42%	17.55%	1.23%	48.89%
povray	done	8.49%	7.25%	10.56%	18.47%	3.20%	50.78%	70.37%	67.22%
omnetpp	done	16.24%	0.08%	21.68%	1.36%	5.63%	27.41%	15.97%	2.17%
xalancbmk	done	16.81%	11.16%	19.36%	22.22%	4.43%	70.85%	7.06%	6.03%
x264_pass1	done	2.29%	0.14%	2.69%	1.01%	3.06%	24.98%	3.70%	14.18%
x264_pass2	done	1.83%	0.11%	2.22%	0.71%	1.96%	36.21%	1.16%	20.88%
x264_seek	done	1.83%	0.12%	2.23%	0.82%	1.89%	32.07%	1.21%	14.73%
gcc_pp_O3	done	8.03%	0.78%	10.65%	4.61%	3.42%	43.50%	8.19%	10.94%
gcc_pp_O2	done	8.27%	0.77%	10.98%	4.65%	3.68%	39.71%	6.79%	9.19%
gcc_smaller	done	9.91%	1.35%	12.74%	3.23%	3.27%	51.24%	4.10%	9.62%
gcc_ref32_O5	done	7.92%	0.97%	10.34%	3.45%	3.54%	54.04%	11.27%	11.98%
gcc_ref32_O3	done	10.27%	1.78%	11.48%	5.26%	9.56%	74.31%	3.44%	15.00%
mcf	done	21.77%	2.59%	27.84%	6.13%	4.77%	72.93%	6.79%	18.85%
perlbench_checkspam	done	1.73%	0.09%	2.73%	0.31%	0.74%	35.71%	3.13%	12.02%
perlbench_diffmail	done	2.75%	0.17%	3.88%	2.31%	0.91%	51.55%	0.00%	21.79%
perlbench_splitmail	done	10.14%	0.03%	14.00%	10.37%	3.67%	6.97%	3.07%	0.24%
bwaves_1	done	7.33%	0.75%	7.74%	0.00%	9.36%	97.66%	18.52%	98.01%
bwaves_2	done	7.29%	0.74%	7.70%	0.00%	9.42%	97.59%	17.34%	99.48%
bwaves_3	done	7.30%	0.81%	7.70%	0.00%	9.35%	97.86%	20.74%	98.31%
bwaves_4	done	7.25%	0.58%	7.67%	0.00%	9.37%	98.43%	14.15%	53.65%
exchange2	done	1.13%	1.29%	1.57%	1.65%	0.43%	1.41%	0.12%	0.63%
fotonik3d	done	8.23%	2.06%	11.58%	0.01%	0.02%	83.12%	2.44%	18.37%
roms	done	10.86%	1.55%	12.52%	0.72%	11.26%	67.88%	13.91%	6.85%
wrf	done	5.68%	5.10%	5.68%	3.73%	5.79%	62.89%	10.21%	17.07%
cam4	done	10.39%	1.80%	11.18%	2.11%	12.00%	56.58%	7.50%	25.29%
pooled		8.19%	2.52%	9.92%	2.88%	5.53%	57.40%	11.76%	8.58%
mean		8.41%	1.71%	10.28%	3.10%	6.57%	53.67%	12.76%	25.22%

Formulas

column	formula	aciklama
spill/ROI	`((spill_load + spill_store) / roi_total_insts) * 100`	ROI komutlarinin yuzde kaci spill load veya spill store
spill ld/load	`(spill_load / roi_load) * 100`	ROI load'larinin yuzde kaci spill reload
spill st/store	`(spill_store / roi_store) * 100`	ROI store'larinin yuzde kaci spill store
(spill+csave)/ROI	`((spill_load + spill_store + csave_load + csave_store) / roi_total_insts) * 100`	ayni oran, callee-save save/restore dahil
(spill+csave) ld/load	`((spill_load + csave_load) / roi_load) * 100`	ROI load'larinin yuzde kaci spill reload veya callee-save restore
(spill+csave) st/store	`((spill_store + csave_store) / roi_store) * 100`	ROI store'larinin yuzde kaci spill store veya callee-save save

column	formula	aciklama
L1D	`(L1D misses / L1D accesses) * 100`	Tum load ve store access'lerinin (spill + other) L1D miss rate'i: L1D'nin bu konfigurasyondaki genel performansi.
spill ld L1	`(spill load L1D misses / spill load L1D accesses) * 100`	Spill load'larin yuzde kaci L1D'de miss oldu. Genelde dusuktur: spill slot'u kisa sure once yazildigi icin cache line cogu zaman hala L1D'dedir.
other ld L1	`((load misses - spill load misses) / (load accesses - spill load accesses)) * 100`	Other load'larin (spill olmayan load'lar) L1D miss rate'i. spill_ld_L1miss ile yan yana bakilir: spill load'lar normal veriden daha mi az, daha mi cok miss oluyor?
spill st L1	`(spill store L1D misses / spill store L1D accesses) * 100`	Spill store'larin yuzde kaci L1D'de miss oldu. Cache write-allocate: miss olan bir store once cache line'i L2'den getirir, sonra yazar.
other st L1	`((store misses - spill store misses) / (store accesses - spill store accesses)) * 100`	Other store'larin (spill olmayan store'lar) L1D miss rate'i; spill_st_L1miss ile karsilastirmak icin.
L2	`(L2 misses / L2 accesses) * 100`	L2'ye gelen tum isteklerin (L1D ve L1I miss'leri) yuzde kaci L2'de de miss oldu ve DRAM'e gitti.
spill ld L2	`(spill load L2 misses / spill load L2 accesses) * 100`	L1D'de miss olan spill load'larin yuzde kaci L2'de de miss oldu. Payda sadece L1D'yi miss eden spill load'lardir, butun spill load'lar degil.
spill st L2	`(spill store L2 misses / spill store L2 accesses) * 100`	L1D'de miss olan spill store'larin yuzde kaci L2'de de miss oldu (payda yine sadece L1D miss'leri).

spill = compiler-confirmed register-allocator spill (GCC `-dP`), csave = callee-saved register save/restore (CFI), both inside the ROI only. pooled = sum of numerators / sum of denominators over the completed runs; mean = average of the per-run percentages. Miss rates are per cache access; an L2 rate's denominator is the L1D misses of that access type.